

Sohan Venkatesh

✉ soh.venkatesh@gmail.com [in sohan-venkatesh](https://www.linkedin.com/in/sohan-venkatesh) [sohv](https://github.com/sohv) [Website](#) 📞 9611768593

Education

Manipal Institute of Technology Bengaluru

Bachelor of Technology in Computer Science (Specialization in AI)

Expected August 2026

GPA: 8.4/10.0

Experience

Precog-at-IIITH

Research Intern

May 2025 – Present

Hyderabad, India

- Investigated how narrow fine-tuning influenced the LLM personality traits through TRAIT benchmark evaluation.
- Fine-tuned Qwen2.5 and Qwen3 on OpenAI's personality datasets, identifying misaligned behavior on Big-Five and Dark Triad traits. Models released on [HuggingFace](#) for reproducibility.
- Performed persona vector analysis to extract steering vectors from the fine-tuned model hidden states and applied these vectors to control base model personality traits.

Fellowship.ai

Data Science Fellow

July 2024 – September 2024

San Francisco, USA

- Designed a Jupyter Copilot extension to assist data scientists with coding, debugging, and testing, improving efficiency by **30%**.
- Engineered CrewAI agents for data preprocessing and model training using GPT-4, reducing latency by **30%** and enhancing code readability by **50%**.

Infosys

Project Intern

May 2024 – July 2024

Bengaluru, India

- Built an ML model using PyCaret and Scikit-learn to predict obesity risk across multiple categories, achieving **83%** accuracy and **82%** precision.
- Applied Pandas and Matplotlib for data analysis, streamlining model training by **30%** and developed a Streamlit web app for real-time prediction.

Selected Projects

Do LLMs really understand causality? | *Research study* 🌐

- Evaluated causal discovery algorithms on real-world and synthetic datasets using SHD, precision, recall and f1-score.
- Benchmarked LLM-simulated causal graphs against algorithmic outputs, showing that LLMs mimic causal patterns but fail to capture true underlying structures.

Not All Thoughts Are Equal | *Research Study*

- Developed **CoT flip test** to identify critical reasoning steps in LLMs and studied the mechanistic representation of reasoning via residual stream analysis.
- Applied activation patching to successfully restore **60%** of correct answers by targeting critical activation patterns in the model layers.

The Holistic Interpretability | *PyTorch, Scikit-learn, MNIST* 🌐

- Implemented ablation, attribution, and mechanistic interpretability techniques for neural network analysis in MNIST.
- Built causal tracing framework analyzing neuron correlations across **128-dimensional** hidden layers to trace information flow through circuits.

Multi Code Review and Generation (MCRAG) | *OpenAI, Gemini, FastAPI* 🌐

- Built automated multi-LLM pipeline for code review and generation using GPT-4o, Gemini 2.5-pro, and Deepseek R1, reducing manual review time by **30%**.
- Improved code quality by **50%** using LLM critic feedback and consensus-based refinement.

nanoKimi | *Kimi-K2, Muon, Latent attention* 🌐

- Implemented Kimi-K2 architecture with Mixture-of-Experts, achieving **18%** faster inference than standard attention.
- Developed custom Muon optimizer and latent attention mechanisms, reducing training memory usage by **30%**.

Achievements

- Awarded **\$1,000 Cohere Lab Catalyst Grant** to build an AI-powered multilingual resume evaluation tool.
- Selected among 3000 students (**5%** acceptance) for Amazon ML Summer School, an intensive ML program.
- Awarded AWS AI & ML Scholarship, selected among 1500 students worldwide (**2%** acceptance).
- Worked on prompt optimization for hallucination reduction in reasoning models as part of EleutherAI's Summer of AI Research program.
- Awarded **Cerebras Fellowship** to work on faster inference in 'Multi Code Review and Generation'.

Technical Skills

Programming Languages: Python, JavaScript, C, SQL, $\LaTeX$

Machine Learning & AI: PyTorch, Scikit-learn, HuggingFace Transformers, PEFT, OpenAI, Gemini, Ollama, Unsloth, Pandas, NumPy, Matplotlib, OpenCV, Neo4j, spaCy, LangChain, LlamaIndex

Full-Stack Development: React, Next.js, Flask, FastAPI, PyTest, MongoDB, MySQL, SQLite

Developer Tools: AWS, Azure, Docker, Git, Jupyter Notebook, VS Code